NRES 746 – Week 2, Lecture 1: R Syntax and Function Adjustments

Name: ____

Instructions: Work alone on Parts 1-2 (about 10-12 minutes total). Then get together with your neighbor(s) to compare answers.

Part 1: Find the bug

Each snippet below has one bug. None of them are typos in the strict sense – the R is syntactically valid enough to run (or at least to try to run). Circle the bug and write your fix directly on the snippet.

1a.

```
ricker <- function(a, b, x) {
  a * x * exp(-b * x)
}

# Evaluate the curve at x = 10, using a = 2, b = 0.5
ricker(10, 2, 0.5)
```

1b.

```
a <- 2; b <- 0.3
curve(a * x * e^(-b * x), from = 0, to = 15)
```

1c.

```
ricker <- function(x, a, b = 0.2) a * x * exp(-b * x)
avals <- c(0.5, 1, 1.5)
peak_vals <- sapply(avals, ricker(), x = 5)
```

Part 2: Write it yourself

The base function $y = \sin(x)$ has period 2π, ranges from -1 to 1, and increases fastest right where it crosses its midline ($y = 0$) heading upward – at $x = 0$ (and every 2π after that).

2a. Write the equation $y(t)$ (`t` = day of year, 1-365), built from sin, whose output:

- repeats with a period of exactly 1 year (365 days)
- ranges from 0 to 10
- increases fastest at $t = 91$ (April 1)

$y(t) =$ ____

2b. Now translate your equation into an R function `seasonal <- function(t) { ... }`.

2c. Name a real natural phenomenon this function could plausibly approximate, and justify your choice in a sentence or two.